INCH-POUND

MS14040B
W/AMENDMENT 1
25 February 2021
SUPERSEDING
MS14040B
29 June 2007

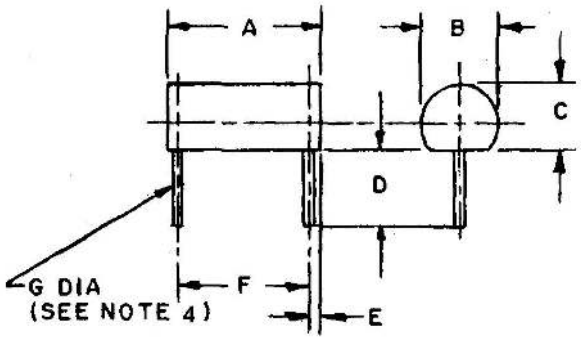
MILITARY SPECIFICATION SHEET

COILS, RADIO FREQUENCY, MOLDED, FIXED, SUBMINIATURE, (IRON CORE), TYPES LT10K097 TO LT10K107 INCLUSIVE

Inactive for new design after 25 July 1983.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-15305.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A	.694 (17.63)	.756 (19.20)
B	.204 (5.18)	.266 (6.76)
C	.189 (4.80)	.251 (6.38)
D	.250 (6.35)	.374 (9.50)
E	.051 (1.30)	.071 (1.80)
F	.590 (14.99)	.610 (15.49)
G	.023 (0.58)	.027 (0.69)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents, in parentheses, are given for general information only.
3. These coils are intended to be mounted by the body.
4. Solderable/weldable lead wire, AWG number 22.

FIGURE 1. Dimensions and configuration.

AMSC N/A

FSC 5950

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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Style: LT10.

Grade: 1.
Class: A.

Weight: 1.8 grams, maximum.

Operating temperature range: -55°C to +105°C.

Ambient temperature: 90°C maximum.

Temperature rise: 15°C maximum.

Terminal pull: 5 pounds minimum.

Altitude: 70,000 feet.

Shock, specified pulse: MIL-STD-202-213, test condition I, is applicable.

Dielectric withstanding voltage:

At sea level: MIL-STD-202-301, test voltage 1,000 V rms for a minimum of 60 seconds.

At reduced barometric pressure: MIL-STD-202-105, test condition C, test voltage 200 V rms for a minimum of 60 seconds.

Electrical characteristics: See table I and table II.

Inductance: See table I.

Q values: See table I.

Self-resonant frequency (SRF): See table I.

DC resistance (DCR): See table I.

Part or Identifying Number (PIN): MS14040- (dash number from table I).

VERIFICATION:

Quality assurance provisions:

Qualification Inspection: Not applicable for this standard.

Conformance Inspection: Group A and group B tests of MIL-PRF-15305 shall be applicable.

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TABLE I. Electrical characteristics (initial).

Dash number 1/	Type designation 2/	Superseded MS PIN	Inductance (μH) ±10%	Q (min)	Test frequency (MHz)	SRF min (MHz)	DC resistance (ohms)	Rated dc current (mA)
-1	LT10K097	MS18100-10	5.60	35	7.9	50	.15	760
-2	LT10K098	MS18100-11	6.80	35	7.9	40	.20	660
-3	LT10K099	MS18100-12	8.20	35	7.9	38	.25	575
-4	LT10K100	MS18100-13	10.0	35	7.9	34	.30	545
-5	LT10K101	MS18100-14	12.0	40	2.5	30	.50	410
-6	LT10K102	MS18100-15	15.0	40	2.5	28	.60	380
-7	LT10K103	MS18100-16	18.0	45	2.5	25	.80	330
-8	LT10K104	MS18100-17	22.0	50	2.5	21	1.20	264
-9	LT10K105	MS18100-18	27.0	55	2.5	19	1.40	248
-10	LT10K106	MS18100-19	33.0	60	2.5	17	1.90	214
-11	LT10K107	MS18100-20	39.0	60	2.5	16	2.00	198

1/ The dash number added to MS military standard number constitutes the MS PIN; for example MS14040-1.

2/ The decrease in maximum operating temperature from 125°C to 105°C does not downgrade these coils but assures satisfactory operation at 105°C for a minimum of 2,000 hours of life rather than a shorter period of operation at 125°C.

TABLE II. Electrical characteristics (final). 1/

Inspection group	Allowable variation from Initial measurement		Allowable percent from specified minimum value in electrical characteristics (initial) table	
	Inductance (percent)	DC resistance	Self-resonant frequency	Q
Qualification inspection				
Group II	±2	---	---	-10
Group III	±5	±(3% +.001 ohm)	-8	-10
Group IV 2/	±5	±(2% +.001 ohm)	-10	-15
Conformance inspection group C				
Subgroup I	±2	---	---	-10
Subgroup II 2/	±5	±(2% +.001 ohm)	-10	-15
Subgroup III	±5	±(3% +.001 ohm)	-8	-10

1/ Test fixture allowance of +.01 μH shall be added to all change in inductance limits ±(_ percent +.01 μH).

2/ The polarizing voltage during the moisture resistance tests Is applied with the positive lead connected to the coil terminals tied together, and the negative lead connected to the metal strap.

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-15305 this document references:

MIL-STD-202-105 MIL-STD-202-213 MIL-STD-202-301

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Review activities:
Army – AR, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19

Preparing activity:
DLA – CC

(Project 5950-2021-002)

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